P510/2
PHYSICS
(Practical)
Paper 2
July/Aug. 2026
3 hours

GLORISO EXAMINATIONS BOARD (GEB)-KAMPALA

SECONDARY SCHOOLS JOINT MOCK ASSESSMENTS, 2026

Uganda Advanced Certificate of Education

PHYSICS

(Practical)

Paper 2

3 hours

INSTRUCTIONS TO CANDIDATES:

- ✓ *This examination paper consists of* **two** *compulsory items.*
- ✓ *Use* **blue** *or* **black** *ink. Any work done in pencil* **except** *graphs and diagrams will* **not** *be scored.*
- ✓ *Scores are awarded mainly for clear record of the observations actually made, for their suitability, accuracy and for the use made of them.*
- ✓ *Details on the question paper should not be repeated in the responses.*
- ✓ *Graph paper is provided.*
- ✓ *Silent non-programmable scientific calculators may be used.*

Item 1.

A district local government plans to construct a wooden footbridge across a small stream to improve access to a nearby health Centre and market. The bridge will be used daily by pedestrians carrying agricultural produce and other light loads. During the rainy season, the bridge may experience additional loading due to water accumulation and increased traffic. A consignment of timber has already been delivered to the construction site, but the supplier did not provide information about its mechanical properties. For safety purposes, the district engineer recommends that timber suitable for the bridge should have a Young's modulus of approximately 5.0×10^9 Pa. Before construction begins, the engineer requires experimental evidence to determine whether the supplied timber possesses adequate stiffness and will not undergo excessive bending when subjected to loads. You have been provided with a sample of the timber and other laboratory apparatus.

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Task:

Plan and carry out an investigation to verify whether the timber is suitable for bridge construction.

Item 2.

An NGO distributed portable emergency lanterns to communities in flood-prone areas to improve lighting during power outages. Shortly after distribution, many users complained that the lanterns became dim after only a short period of use, making them unreliable during emergencies. The supplier maintained that the dry cells used in the lanterns met the required specifications and stated that a good-quality cell should have an internal resistance of approximately 0.8 Ω. Since excessive internal resistance reduces the current supplied to a device, the organization became concerned about the quality of the cells supplied. To establish the cause of the problem, the organization appointed a team of technicians to investigate the electrical properties of the cells. The technicians were instructed to determine the internal resistance of the

cells using two methods, one of which should involve the use of a potentiometer slide wire, a Centre-zero galvanometer and the dry cells under investigation, and then obtain an average value for improved reliability of the results. You have been provided with a sample of the dry cells and other laboratory apparatus.

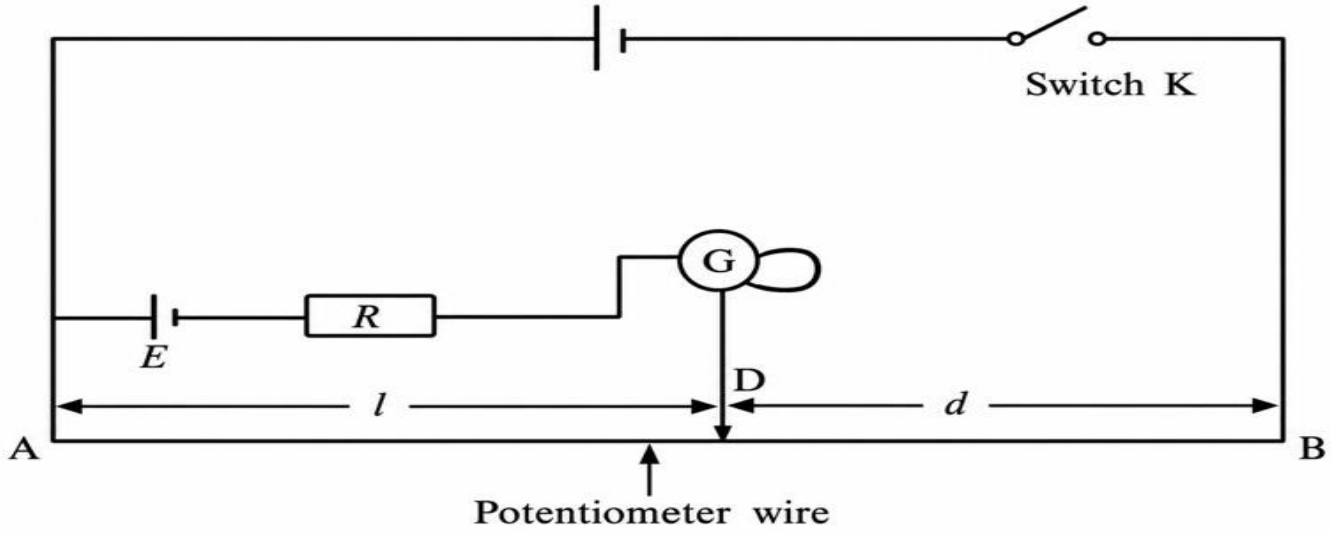
Task:

Follow the instructions provided and carry out an investigation to determine whether the supplied dry cells satisfy the required quality standards.

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Part I

(a) Connect the circuit as shown in the figure below.



(b) Switch on the switch, K

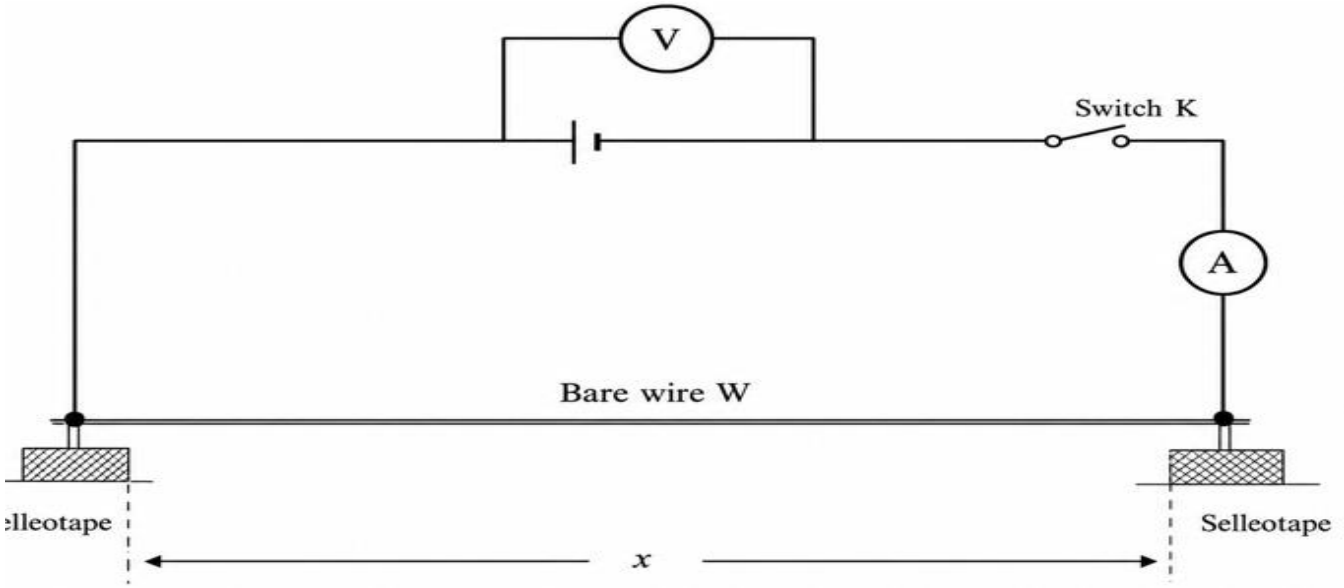
(c) By moving the slide contact along AB, obtain a point D on AB where the galvanometer shows no deflection.

(d) Measure and record the length, l in cm.

(e) Calculate the internal resistance from the expression: $r_1 = (\frac{100}{l} - 1)$

Part II

(a) Connect the circuit as shown below.



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(b) Measure the length, X= 10.0 cm

(c) Close the switch, K

(d) Measure and record the Voltmeter and Ammeter reading, *V and I* respectively.

(e) Open the switch, K

(f) Repeat the procedure (b) to (e) for X=20.0, 30.0, 40.0 and 50.0 cm respectively.

(g) Record results in a suitable table.

(h) Plot a graph of V against I

(i) Calculate the slope, S of the graph.

(j) Determine the internal resistance of the cell using the expression $r_2 = S$

(k) Provide feedback whether *the supplied dry cells satisfy the required quality standards.*

"GOOD LUCK"

****THE END****